LESSON PLAN

AVIONICS AND ELECTRONIC WARFARE (13)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	01
Total No. of Lesson Plans	25	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	DC power supply systems, rectified power supplies
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	DC power supply systems, rectified power supplies.
5. Specific Objectives	13.1.1.1 Trainee will be able to explain the A/C power systems.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of schematic diagrams; Instructor supervision of Q&A; session
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of a DC power supply?
(b) How does an aircraft AC power system differ from DC?
(c) Why is rectification necessary in avionics systems?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• AC power generation principles • Transformer operation basics • Rectifier circuit configurations • Filtering components function • Regulation mechanisms	L & Ds	PPT, IB	15 min
(b)	• Half-wave vs full-wave rectification • Bridge rectifier topology • Capacitor input filtering • Voltage ripple calculation • Heat dissipation in rectifiers	L & Ds	PPT, PDF	20 min
(c)	• System integration in aircraft • Failure modes analysis • Maintenance requirements • Safety protocols • Troubleshooting steps	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the role of a bridge rectifier in converting AC to DC.
(b) Describe how a capacitor filter reduces voltage ripple in a power supply.

12. Home Assignment and Follow Up	Prepare a brief note on the constructional details of generators and engine coupling to be discussed in LP 02.

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Generator, engine coupling, cooling of generators, constructional details of brushes types of brushes and causes of brush wear and spark suppression techniques.	Refer to the master syllabus reference list	LP 02

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min